

CO5-AT-2 SHORT ANSWERS TEST

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Course Code: CSA 0307
Course Name: Data Structures

Question 1:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the course name "SIMATS Online Programming Lab", and the user's name "Dhorakonda Suresh" with a dropdown arrow. The main title is "CO5-AT2-Short Answers Test". On the left, a list of questions (Q1 to Q7) is shown, each with a green checkmark and "2 marks". The selected question is "Q1: Define a graph." The answer field contains the text "A graph is data struetue consisting of vertices and edges connecting pair of vertices". Below the answer field are buttons for "Save Answer" and "Next".

Q1: Define a graph. 2 marks

Define a graph.

Your Answer

A graph is data struetue consisting of vertices and edges connecting pair of vertices

Save Answer Next

Question 2:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the course name "SIMATS Online Programming Lab", and the user's name "Dhorakonda Suresh" with a dropdown arrow. The main title is "CO5-AT2-Short Answers Test". On the left, a list of questions (Q1 to Q7) is shown, each with a green checkmark and "2 marks". The selected question is "Q2: What are the major types of graphs?". The answer field contains the text "There are two major type of graphs 1.Directed graph 2.Undirected graph". Below the answer field are buttons for "Save Answer" and "Next".

Q2: What are the major types of graphs? 2 marks

What are the major types of graphs?

Your Answer

There are two major type of graphs
1.Directed graph
2.Undirected graph

Save Answer Next

Question 3:



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QUESTIONS

Q1 Define a graph.
2 marks

Q2 What are the major types of graphs?
2 marks

Q3 Differentiate between directed and undirected graphs.
2 marks

Q4 What is a weighted graph?
2 marks

Q5 What is a topological sort?
2 marks

Q6 A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks

Q7 An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
2 marks

7/7 answered

Q3 Differentiate between directed and undirected graphs. 2 marks

Differentiate between directed and undirected graphs.

Your Answer

Directed graphEdges have specific direction
Undirected graphEdges have no direction

Save Answer Next →

Question 4:



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QUESTIONS

Q1 Define a graph.
2 marks

Q2 What are the major types of graphs?
2 marks

Q3 Differentiate between directed and undirected graphs.
2 marks

Q4 What is a weighted graph?
2 marks

Q5 What is a topological sort?
2 marks

Q6 A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks

Q7 An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
2 marks

7/7 answered

Q4 What is a weighted graph? 2 marks

What is a weighted graph?

Your Answer

A graph in which each edge has an association value or weight such cost , distance or time

Save Answer Next →

Question 5:



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QUESTIONS

Q1
 Define a graph.
 2 marks

Q2
 What are the major types of graphs?
 2 marks

Q3
 Differentiate between directed and undirected graphs.
 2 marks

Q4
 What is a weighted graph?
 2 marks

Q5
 What is a topological sort?
 2 marks

Q6
 A directed graph has the dependencies $A \rightarrow C, B \rightarrow C, C \rightarrow D, B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
 2 marks

Q7
 An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
 2 marks

7/7 answered

Q4 What is a weighted graph?
2 marks

What is a weighted graph?

Your Answer
 A graph in which each edge has an association value or weight such cost , distance or time

Save Answer
Next →

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QUESTIONS

Q1
 Define a graph.
 2 marks

Q2
 What are the major types of graphs?
 2 marks

Q3
 Differentiate between directed and undirected graphs.
 2 marks

Q4
 What is a weighted graph?
 2 marks

Q5
 What is a topological sort?
 2 marks

Q6
 A directed graph has the dependencies $A \rightarrow C, B \rightarrow C, C \rightarrow D, B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
 2 marks

Q7
 An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
 2 marks

7/7 answered

Q6 A directed graph has the dependencies $A \rightarrow C, B \rightarrow C, C \rightarrow D, B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks

A directed graph has the dependencies $A \rightarrow C, B \rightarrow C, C \rightarrow D, B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

Your Answer
 A-C; A apperas before C
 B-C; B apperas before C
 C-D; C apperas before D
 B-E; B apperas before E
 E-F; E apperas before F
 Topological order ;B,A,C,D,E,F

Save Answer
Next →

Question 7:

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QUESTIONS

Q1 ✓
Define a graph.
2 marks

Q2 ✓
What are the major types of graphs?
2 marks

Q3 ✓
Differentiate between directed and undirected graphs.
2 marks

Q4 ✓
What is a weighted graph?
2 marks

Q5 ✓
What is a topological sort?
2 marks

Q6 ✓
A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks

Q7 ✓
An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
2 marks

Q6 ✓ **A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.** 2 marks

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

Your Answer

A-C; A appears before C
B-C; B appears before C
C-D; C appears before D
B-E; B appears before E
E-F; E appears before F
Topological order ;B,A,C,D,E,F

[Save Answer](#)[Next →](#)